

Embeddings, Vectors, Similarity Search & Vector Search

Topic 1 - What Are Embeddings?

An Embedding is a numerical representation of text.

Example:

Azure OpenAI

becomes:

[0.12, 0.98, -0.45, 0.11, ...]

AI converts words into numbers.

Why?

Computers don't understand text.

Computers understand:

Numbers

Embeddings allow AI to understand:

Meaning

instead of:

Exact Words

Topic 2 - Why Embeddings Matter

Suppose user asks:

How do I configure Service Bus?

Document contains:

Azure Service Bus Setup Guide

Keyword search:

✗ Might fail

Embedding search:

✓ Understands both mean the same topic

Topic 3 - What Is A Vector?

A vector is simply:

List Of Numbers

Example:

[0.11, 0.22, -0.44, 0.89]

Every document becomes a vector.

Every query becomes a vector.

Topic 4 - Similarity Search

Goal:

Find vectors that are close together.

Example:

Server Issue

and

Application Failure

may produce similar vectors.

The search engine finds:

Relevant Information

instead of exact matches.

Topic 5 - Traditional Search vs AI Search

Traditional Search

Keyword Match

Search:

car

Matches:

car

Doesn't understand:

automobile

vehicle

transport

AI Search

Search:

car

Understands:

vehicle

automobile

transport

Because embeddings capture meaning.

Topic 6 - Azure OpenAI Embeddings

Azure OpenAI provides embedding models that convert content into vectors. Microsoft documentation describes embeddings as floating-point vectors representing semantic meaning, where similar text produces nearby vectors.

Architecture

Document



Embedding Model



Vector

Example

Project Document



Azure OpenAI Embeddings



Vector

Stored for future search.

Topic 7 - Azure AI Search Vector Search

Azure AI Search supports vector search over numeric representations of content and can be used for similarity search, hybrid search, multilingual search, and more.

Flow

Documents



Embeddings



Azure AI Search



Vector Index

Topic 8 - Hybrid Search

Hybrid Search combines:

Keyword Search

+

Vector Search

Azure AI Search supports this pattern for enterprise search scenarios.

Example

Query:

Approve Pull Request

Document:

Code Review Process

Hybrid search helps retrieve the right content even when wording differs.

Topic 9 - How RAG Works

This is the first real introduction.

Problem

GPT doesn't know your internal documents.

Solution

User Question



Embedding



Vector Search



Relevant Documents



GPT Model



Answer

This is:

RAG

Topic 10 - Enterprise Example

Imagine:

IDD Tech Knowledge Assistant

Documents:

- Support Guides
- Project Documents
- Architecture Docs
- Runbooks

User Question

How does SensGrid authentication work?

System Flow

Question

↓

Embedding

↓

Vector Search

↓

Relevant Documents

↓

GPT Model

↓

Answer

Part 2 - Understanding The Architect View

As a Solution Architect, think:

Not

What is an embedding?

Think

Where are embeddings generated?

Where are embeddings stored?

How is security applied?

What is the cost?

How is retrieval optimized?

How is monitoring done?

Architecture View

Documents



Chunking



Embedding Generation



Vector Storage



Retrieval Layer



GPT Model



Answer

Part 3 - Mini Hands-On

No coding required today.

Draw this in Draw.io

Documents



Azure OpenAI Embeddings



Azure AI Search



Vector Search



.NET API



Azure OpenAI GPT



User Response

Save:

Embeddings.drawio

Embeddings.png

Assignment

Answer:

Question 1

What is an embedding?

Question 2

Why do embeddings improve search?

Question 3

What is a vector?

Question 4

Why is vector search important?

Question 5

Why is hybrid search useful?

Question 6

How do embeddings support RAG?